Ex.no : 4

Date : 12.9.26

Routing at Network Layer.

a) Simulate static Routing Protocol configuration using Cisco Packet Tracer.

Aim :

To simulate and configure static Routing Protocol using Cisco Packet Tracer and observe packet routing between different networks.

Procedure :

1) Open Cisco Packet Tracer.
2) Place three routers, two switches and four PCs.
3) Connect the devices using appropriate Ethernet cable.
4) Enable the router interface using the no shutdown command.
5) Configure a primary routes on the routers using ip route command
6) Configure a primary route and backup route using different administrative distance.
7) Verify using routing table show ip route static
8) Test connectivity between PCs using the ping command.
9) Use tracert <destination-Ip> to observe destination.
10) Simulate the failure the main route and verify that the backup route is used.

Result :

Thus the static Routing was successfully simulated and configured using Cisco Packet Tracer through primary and backup routes was verified.

5) Simulate RIP using Packet Tracer

Aim:

To simulate RIP using Cisco packet tracer and observe dynamic routing between different network.

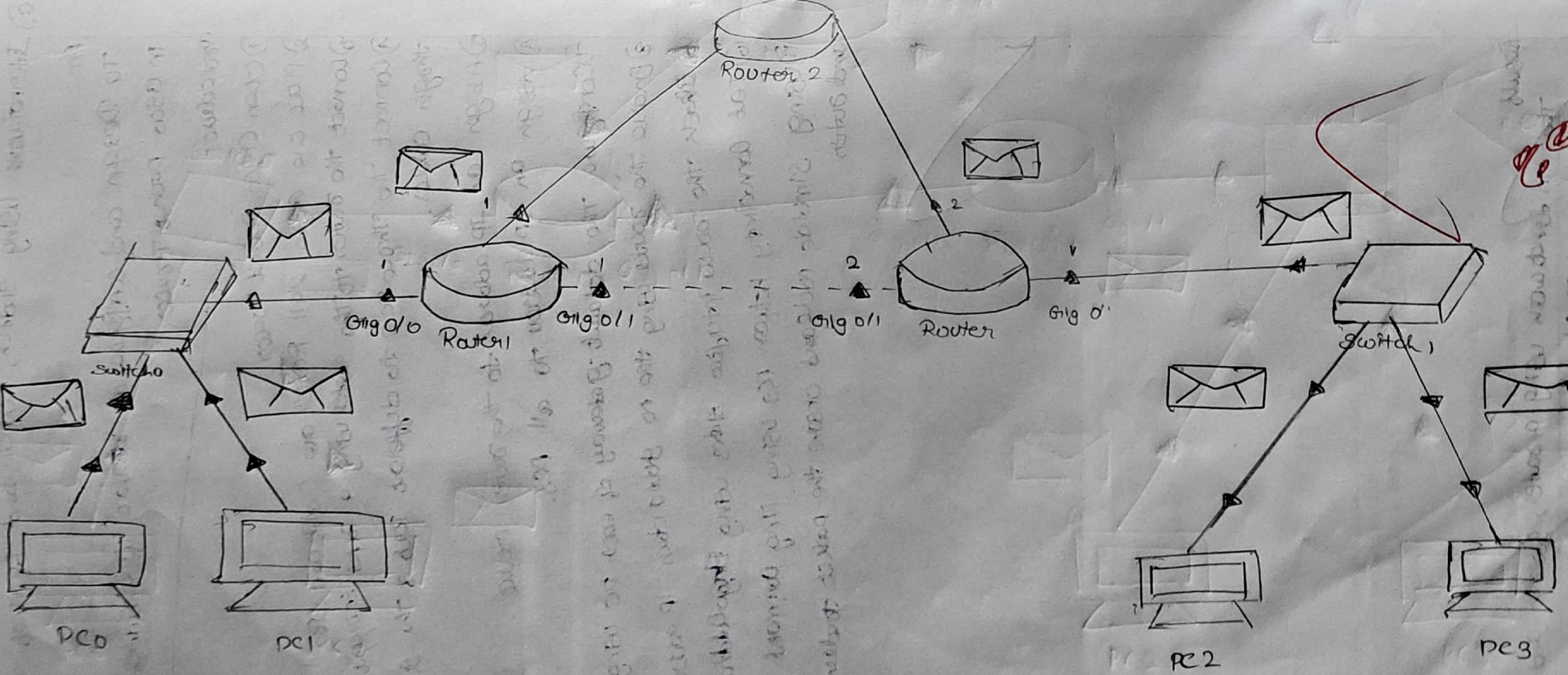
Procedure:

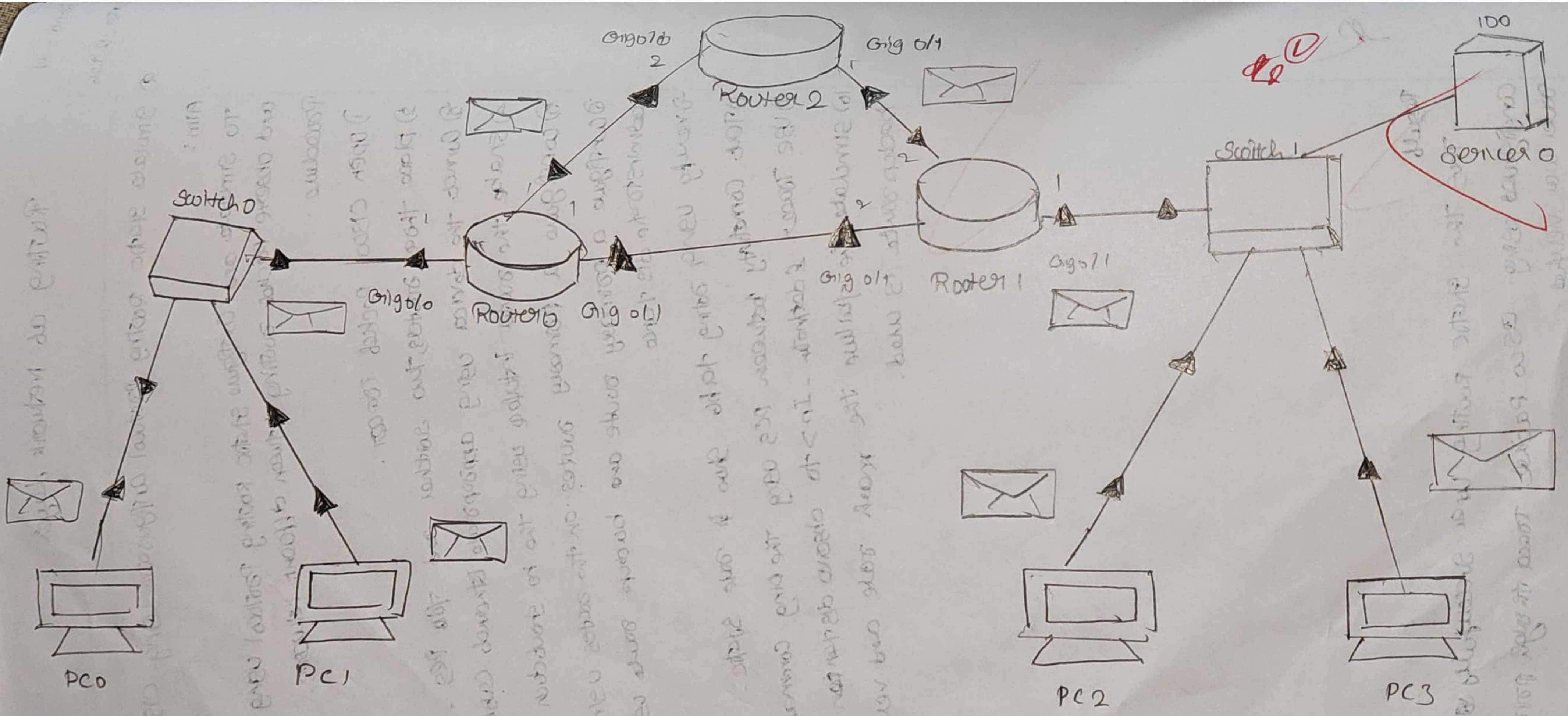
1) Open Cisco Packet tracer.

2) Place three router and Two PCs
3) Connect those routers in a triangle topology.
4) Connect the PC0 to Router 0 and PC1 to Router 1.
5) Configure the IP address of all Router interfaces.
6) Enable the interface using no shutdown.
7) Configure RIP on each router using router rip.
8) Advertise the directly connected networks using the network command.
9) Verify the routing table using show ip route.
10) Test connectivity between PC0 and PC1 using.
11) Use Trace to observe the route selected by RIP.
12) Disconnect one route and observe the RIP automatically selected another available route.

Result:

Thus, RIP was successfully simulated using Cisco Packet Tracer and Dynamic route select and automatic rerouting were verified.


Router 2
1
2
1
Gig 0/0
Router1
1
Gig 0/1
2
Gig 0/1
Router
Gig 0/
Switch0
Switch1
PC0
PC1
PC2
PC3


Gig0/0
2
Router2
Gig 0/1
1
2
1
Switch0
1
Gig0/0
Router0
Gig 0/1
1
2
Gig 0/1
Router 1
Gig0/1
Switch 1
100
Server0
PC0
PC1
PC2
PC3